

Proposal for Canvas User Feedback System

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Introduction

Canvas is a widely used Learning Management System (LMS) that serves millions of students and educators across educational institutions worldwide. While Canvas provides essential functionality for course management, assignment submission, and instructor-student communication, there is significant room for improvement in how user feedback is collected and processed. This project aims to develop an integrated feedback system that will enhance user experience by providing a streamlined mechanism for reporting issues, suggesting improvements, and rating various aspects of the platform's usability.

Project Description

The proposed Canvas User Feedback System will be a demonstration prototype that showcases how user feedback could be collected effectively in a learning management system environment. While not directly integrated with Canvas, the prototype will demonstrate the potential implementation of feedback collection through multiple channels while maintaining a minimal impact on the user's primary tasks. The project consists of 3 main components:

1. *Feedback Collection Interface* - A user-friendly interface offering multiple interaction styles (direct manipulation, menu selection, and form fill-in) to accommodate different user preferences and task contexts.
2. *Usability Testing* - Comprehensive testing of the feedback system's effectiveness using both expert reviews and user testing.
3. *Survey Design* - Development of well-structured surveys with appropriate menu organization and form fill-in techniques based on established HCI principles.

Key features of the feedback system include:

- An always-accessible feedback button integrated into the Canvas navigation bar
- Context-aware feedback forms that adapt based on the user's current location within the platform
- A structured categorization system for different types of feedback (bug reports, feature requests, usability issues)
- Real-time sentiment analysis of feedback content to prioritize urgent issues

- A dashboard for users to track the status of their submitted feedback
- Support for multiple interaction styles to accommodate varying user preferences and contexts

Objectives

The primary objectives of this feedback system are to:

1. Improve the overall user experience of Canvas by providing an intuitive and accessible mechanism for users to voice their concerns and suggestions
2. Reduce the friction in reporting technical issues and usability problems by implementing a context-aware feedback collection system
3. Create a structured approach to gathering and categorizing user feedback that will facilitate more efficient processing and resolution of issues
4. Establish a transparent feedback loop between users and platform administrators
5. Generate actionable insights from collected feedback to guide future platform improvements

HCI Techniques and Design

We will incorporate several established HCI principles and methodologies such as:

User-Centered Design (UCD):

- Conducting user interviews and surveys to understand feedback submission preferences
- Creating user personas to represent different stakeholder groups (students, instructors, administrators)
- Developing user journey maps to identify optimal feedback collection points
- Implementing iterative design processes based on user testing results

Interface Design Principles:

- Following Shneiderman's Eight Golden Rules of interface design:
 - Strive for consistency in action sequences, terminology, and layout
 - Cater to universal usability by providing features for novice and expert users

- Offer informative feedback for every user action
- Design dialogs to yield closure so users know when tasks are completed
- Prevent errors through careful design and validation
- Permit easy reversal of actions to encourage exploration
- Support internal locus of control so users feel in command
- Reduce short-term memory load with simple displays and consolidated multiple page designs
- Ensuring accessibility compliance with WCAG 2.1 guidelines
- Maintaining consistency with Canvas's existing design language
- Applying Gestalt principles for visual organization and cognitive processing

Interaction Style Selection:

- Employing multiple interaction styles based on task requirements:
 - Direct manipulation for intuitive drag-and-drop feedback components
 - Menu selection for structured categorization of feedback types
 - Form fill-in for detailed feedback submission with appropriate field organization
- Designing interaction flows using transition diagrams to ensure logical progressions
- Creating appropriate menu hierarchies for feedback categorization
- Applying principles of natural language for error messages and instructional content

Form Design Guidelines:

- Implementing clear field labels and logical grouping of related form elements
- Using appropriate form controls (radio buttons, checkboxes, text areas) based on input type
- Providing format-specific fields for structured data entry when appropriate
- Creating well-organized dialog boxes with consistent layouts and standard buttons
- Including clear completion signals and validation feedback

Tools and Technologies

The development of the feedback system will utilize the following tools and technologies:

Development Stack:

- Frontend: HTML, CSS, and JavaScript for the prototype interface
- Backend: Simple server setup (if needed) for demonstration purposes
- Local storage or simple database for prototype feedback management

Interaction Styles Implementation:

- Direct manipulation interfaces using drag-and-drop libraries
- Menu selection frameworks with appropriate categorization hierarchies
- Form fill-in components with validation and responsive design
- Command language options for advanced users
- Speech recognition integration for accessibility (if appropriate)

Design Tools:

- Figma for interface design and prototyping
- Maze for remote usability testing
- Rapid contextual design methodologies for efficient prototyping

Information Architecture:

- Task-related organization strategies for menu structures
- Hierarchical command structures for navigation
- Combined multiple menu approaches (linear sequences and simultaneous menus)
- Consistent menu layout guidelines for internal consistency

Analytics Tools:

- Google Analytics for user behavior tracking
- Hotjar for heatmap analysis and session recording
- Natural Language Processing (NLP) tools for feedback content analysis

Evaluation and User Testing

The evaluation strategy will employ a mixed-methods approach to ensure comprehensive testing of the feedback system, drawing from established HCI evaluation techniques like:

Expert Reviews:

- Conducting heuristic evaluations using Nielsen's usability principles
- Performing cognitive walkthroughs to identify potential user difficulties
- Implementing consistency inspections to ensure uniform design patterns
- Using a “human-thinking approach” to relate design to users' conceptual models

Usability Testing:

- Conducting moderated usability testing sessions with 3-4 participants per user group
- Implementing think-aloud protocols during testing sessions
- Recording task completion rates and time-on-task metrics
- Gathering qualitative feedback through post-test interviews
- Measuring five key usability factors: time to learn, speed of performance, rate of errors, retention over time, and subjective satisfaction

Iterative Design Evaluation:

- Using low-fidelity prototypes for early concept testing
- Employing high-fidelity prototypes for detailed interaction testing
- Conducting continuous evaluation throughout the development process

Survey Instruments:

- Deploying the Questionnaire for User Interaction Satisfaction (QUIS)
- Creating custom surveys to assess specific aspects of the interface
- Collecting user-subjective impressions about representation of task domain objects and actions
- Measuring user satisfaction with various interaction styles (direct manipulation, menu selection, form fill-in)

Data-Driven Approach:

- Using visualizations to present findings to stakeholders
- Involving stakeholders in the evaluation and refinement process
- Establishing metrics for measuring feedback system effectiveness

Project Timeline

<i>Time Period</i>	<i>Activities</i>
<i>Weeks 1 - 2:</i> Research & Requirements	• Initial research and user interviews • Requirements gathering and analysis • Project planning and resource allocation • User persona development • Creation of initial design concepts
<i>Week 3:</i> Initial Design & Prototyping	• Interface design and wireframing • User flow mapping • Development of low-fidelity prototypes • Expert review of initial concepts • Prototype refinement based on expert feedback
<i>Weeks 4 - 5:</i> Interactive Prototype Development	• Frontend prototype development • Implementation of multiple interaction styles • Local data storage setup • Integration of form design principles • Creation of feedback categorization system
<i>Week 6:</i> Initial Testing	• Heuristic evaluation with 3-5 experts • Prototype refinement based on expert feedback • Development of usability testing plan • Recruitment of usability test participants • Finalization of testing materials and metrics
<i>Week 7:</i> User Testing	• Conduct moderated usability testing sessions • Analysis of test results • Documentation of user feedback • Identification of usability issues • Prioritization of required modifications
<i>Week 8:</i> Refinement & Documentation	• Implementation of usability improvements • Documentation of design decisions • Preparation of demonstration materials • Development of presentation content • Finalization of project deliverables
<i>Week 9:</i> Presentation & Delivery	• Final usability verification • Preparation of final report • Development of presentation slides • Prototype demonstration preparation • Project presentation and submission

Expected Outcomes

1. *Improved User Experience:*
 - a. Increased user satisfaction with Canvas platform; measured through improved NPS scores
 - b. Reduced time-to-resolution for reported issues through better categorization and prioritization
 - c. More efficient feedback submission process through carefully designed interaction flows
2. *Enhanced Feedback Quality*
 - a. Higher quality of feedback submissions through structured input forms and contextual prompts
 - b. Improved platform stability through faster identification and resolution of technical issues
 - c. Better-informed product development decisions based on systematic analysis of user feedback
3. *Data-Driven Insights*
 - a. Comprehensive dashboard of user feedback patterns and trends
 - b. Visualization of problem areas within the Canvas interface
 - c. Identification of most requested features and improvements
4. *Design Knowledge*
 - a. Validation of multiple interaction style effectiveness for feedback collection
 - b. Comparative performance metrics for different form designs and menu structures
 - c. Best practices for feedback system design in learning management systems
5. *Demonstration of HCI Principles*
 - a. Practical application of the Eight Golden Rules of interface design
 - b. Successful implementation of user-centered design methodologies
 - c. Effective use of different interaction styles for appropriate tasks
 - d. Validation of form fill-in and menu selection guidelines

Conclusion

The proposed “Canvas User Feedback System” represents a significant step forward in improving the platform's ability to gather and respond to user input. By implementing this system, Canvas will be able to better understand and address user needs, leading to improved user satisfaction and platform quality. The project's success will be measured through concrete metrics including user engagement rates, feedback resolution times, and overall platform satisfaction scores.

Through careful attention to HCI principles and methodologies, combined with robust technical implementation, this feedback system will provide a valuable tool for both users and platform administrators. The systematic approach to gathering and analyzing feedback will ensure that Canvas continues to evolve in alignment with user needs and expectations.